

The binary cross-entropy loss function:

$$L(\alpha) = -[y \ln \alpha + (1-y) \ln (1-\alpha)]$$

α = Predicted probability

y = True label

$$\therefore \frac{d}{d\alpha} L(\alpha) = \frac{d}{d\alpha} [-y \ln \alpha - (1-y) \ln (1-\alpha)]$$

$$= -\left[y \frac{d}{d\alpha} \ln \alpha + (1-y) \frac{d}{d\alpha} \ln (1-\alpha) \right]$$

$$= -\left[y \cdot \frac{1}{\alpha} + (1-y) \cdot \frac{(-1)}{(1-\alpha)} \right]$$

$$= -\left[\frac{y}{\alpha} - \frac{(1-y)}{(1-\alpha)} \right]$$

$$\therefore \frac{d}{d\alpha} L(\alpha) = \frac{1-y}{1-\alpha} - \frac{y}{\alpha}$$

$$= \frac{\alpha(1-y) - y(1-\alpha)}{\alpha(1-\alpha)}$$

$$= \frac{\alpha - \alpha y - y + \alpha y}{\alpha(1-\alpha)}$$

$$= \frac{\alpha - y}{\alpha(1-\alpha)}$$

In neural network,

α = the output of the sigmoid activation function applied to logits z .

$$\alpha = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\therefore \frac{dL}{dz} = \frac{dL}{d\alpha} \cdot \frac{d\alpha}{dz} \quad (\text{according to chain rule})$$

We got,

$$\frac{dL}{d\alpha} = \frac{\alpha - y}{\alpha(1-\alpha)} \quad \text{and,}$$

$$\frac{d\alpha}{dz} = \alpha(1-\alpha) \quad (\text{from our previous task})$$

$$\begin{aligned} \therefore \frac{dL}{dz} &= \frac{dL}{d\alpha} \cdot \frac{d\alpha}{dz} \\ &= \frac{(\alpha - y)}{\cancel{\alpha(1-\alpha)}} \cdot \frac{\cancel{\alpha(1-\alpha)}}{1} \\ &= \alpha - y \end{aligned}$$

$$\therefore \text{Derivative of loss function, } \frac{dL}{d\alpha} = \alpha - y.$$